4. (4 pts) What is the typical charge state of U in UO_2? What can cause it to change?

U+4. Oxygen vacancies/interstitials and substitutional fission products can force the charge state on U atoms to change to maintain charge neutrality.

5. (5 pts) How does the oxygen concentration change across the fuel pellet? How does Mo impact this?

The oxygen concentration typically goes up with burnup and reaches a maximum when the Mo/MoO2 reaction starts to take place. Thus, Mo acts as a buffer to the oxygen potential in the fuel. The oxygen potential decreases near the periphery of the fuel due to oxygen uptake by the Zr cladding.

6. (3 pts) Name three properties that vary as a function of stoichiometry in UO2.

Lattice constant, thermal conductivity, oxygen diffusion, etc.

7. (10 pts) What are the five types of fission products that form in the fuel? List a role each plays in fuel performance.

Soluble oxides: Go into the fluorite sublattice, cause swelling, decrease kth
Insoluble oxides: Form oxide precipitates, cause swelling, typically decrease kth
Noble metals: Form metallic precipitates, increase kth
Volatile gases: Form gas bubbles, cause swelling, form the corrosive environment for SCC
Noble gases: Form gas bubbles, cause swelling, decrease kth

8. (8 pts) List and describe the three stages of fission gas release.

1: Fission gas is produced and diffuses towards the grain boundaries. 2: Grain boundary gas bubbles form and begin to grow and coalesce. 3: Interconnected tunnels of gas bubbles form along grain edges. Where these tunnels reach a free surface the gas is released.

9. (5 pts) What is creep? What is one mechanism of creep (name and physical description)?

Permanent deformation due to a stress which is below the yield stress. Occurs over time and is volume-conserving. Nabaro-Hering: point defect diffusion. Coble creep: grain boundary diffusion. Power law: dislocation climb.

10. (8 pts) What is the high burnup structure? Why does it form? What are some performance aspects of this structure?

The HBS is a highly porous fine-grained microstructure that develops on the outer rim of the fuel pellet at high burnup. It occurs because of self-shielding effects leading to higher powers on the periphery of the fuel pellet, combining with lower temperatures on the outside of the pellet, leading to defect accumulation and an eventual grain refinement. This structure slightly increases thermal conductivity and increases gas retention.

11. (6 pts) Describe the irradiation growth of hcp Zr due to point defects.

Interstitials want to cluster on pyramidal faces of the HCP lattice. This causes an expansion in *a* and a contraction in *c*. With sufficient dose, vacancies cluster preferentially on the basal plane, causing further shrinkage in the *c* axis.

12. (8 pts) Describe how creep impacts the Zr cladding over time in a fuel rod.

Initially have a high coolant pressure which causes creep down of the cladding. This leads to a decrease in the cladding radius and an increase in the cladding length. The cladding creeps down and comes into partial contact with the fuel. Where there is contact, the pellet exerts an outward force on the cladding, causing a radial increase and an axial shrinkage. The pellet also exerts an axial force, leading to local elongation. Where there is no contact, the cladding continues to creep down due to the coolant pressure. Finally, full contact of the cladding and the fuel column occurs. The fuel exerts both radial and axial forces on the cladding, leading to axial shortening and elongation, respectively. This results in a net axial shrinkage of the cladding and a radial expansion.

13. (12 pts) What are the four conditions that must be met for SCC? Briefly describe how each is met in PCI.

Corrosive environment: Volatile fission products diffuse down the temperature gradient and accumulate at the fuel/cladding interface. These fission products can undergo corrosive reactions with the Zr cladding.
Susceptible material: All Zr alloys are susceptible to SCC. Slight modifications in alloying can reduce but not eliminate this susceptibility.
Sufficient stress: Stress is induced by the pellet swelling and exerting an outward force on the cladding, leading to large hoop tensile stresses. Pellet cracks or missing surfaces can increase the local stress.
Sufficient time: SCC requires the buildup of corrosive fission products as well as the breaking of the protective ZrO_2 layer that forms on the inside of the cladding.

14. (5 pts) What role does iodine play in the corrosion of Zr cladding?

Iodine is a volatile fission product which can form ZrI_4 compounds with the Zr cladding. This gaseous species diffuses up the temperature gradient, where it is then converted into $ZrO_2 + I_2$. The ZrO_2 deposits on the crack walls, while the I_2 is free to interact